071306T4 EEN

ELECTRICAL ENGINEERING (POWER OPTION) LEVEL 6

ENG/OS/PO/CR/02/6

INSTALL ELECTRICAL POWER LINES

March/April 2024

Time: 3 Hours

TVET CURRICULUM DEVELOPMENT, ASSESSMENT AND CERTIFICATION COUNCIL (TVET CDACC)

WRITTEN ASSESSMENT

3 HOURS

INSTRUCTIONS TO CANDIDATE:

i. *This paper consists of **TWO** sections: **A** and **B**.*

ii. *Answer **ALL** questions in sections A and any **THREE** in section B in the answer booklet provided.*

iii. *Marks for each question are indicated in brackets.*

iv. *Do not write on this question paper.*

v. *Answer the questions in **English.***

This paper consists of THREE printed pages.

Candidate should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

SECTION A (40 MARKS)

*Answer **all** the questions in this section.*

1. State FOUR systems of power transmission. (4 marks)
2. State FOUR important components of an overhead transmission line. (4 marks)
3. Highlight FOUR advantages of pin-type insulator used in power transmission lines. (4 marks)
4. Explain TWO purposes of suspension insulators as used in high voltage power transmission. (4 marks)
5. Outline THREE methods of improving string efficiency as used in design of transmission lines. (6 marks)
6. Explain TWO methods of reducing corona effect in an overhead transmission line. (4 marks)
7. Define the term "sag" as used in overhead lines. (2 marks)
8. Outline TWO factors which affect the sag of overhead transmission lines. (2 marks)
9. State THREE classifications of overhead transmission lines citing their voltage levels. (6 marks)
10. Outline FOUR properties of conductor material used for transmission and distribution of electric power. (4 marks)

SECTION B (60 MARKS)

Answer ***three*** *questions in this section*

11. (a) Describe the improvement of string efficiency by grading of units and guard ring. (6 marks)

(b) The three bus-bar conductors in an outdoor substation are supported by units of post type insulators. Each unit consists of a stack of 3 pin type insulators fixed one on the top of the other. The voltage across the lowest insulator is 13·1 kV and that across the next unit is 11 kV. Find the bus-bar voltage of the station. (14 marks)

12. (a) Explain THREE properties of line supports in overhead transmission line. (6 marks)

(b) Determine the efficiency and regulation of a 3-phase, 100 km, 50 Hz transmission line delivering 20 MW at a p.f. of 0.8 lagging and 66 kV to a balanced load. The conductors are of copper, each having resistance 0.1 ohm per km, 1.5 cm outside diameter, spaced equilaterally 2 meters between centers. Neglect leakage and use nominal-T, method. (14 marks)

13. (a) Explain THREE importance of transposition in power transmission lines. (6 marks)

(b) Highlight THREE advantages of Bundled - Conductor lines over Single- conductor lines of the same area of Cross section. (6 marks)

(c) Describe FOUR types of tests done on transmission lines. (8 marks)

14. (a) Outline THREE assumptions made when deriving sag equation for equal supports. (6 marks)

(b) A transmission line conductor having a diameter of 19.5 mm weighing 0.85 kg/m. The span is 275 meters. The wind pressure is 39 kg/m^2 of projected area with ice coating of 13 mm. The ultimate strength of the conductor is 8000 kg. Calculate the maximum sag if the factor of safety is 2 and ice weighs 910 kg/m^3. (14 marks)

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